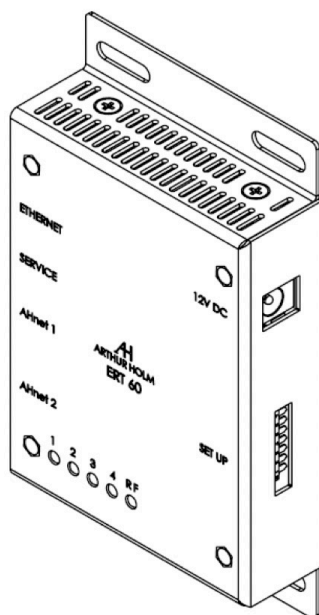




User Guide



ERT

IP to RS422/RS485 interface control

Technology Serving Design

Danish craftsmanship
Mediterranean soul


ARTHUR HOLM

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WELCOME

Thank you for purchasing an ARTHUR HOLM product.

Please, read this installation and operating instructions carefully and keep them in a safe place for future reference.

We remain at your entire disposal if you have any suggestions.

Henrik Holm

General Manager

hholm@albiral.com

ABOUT US

The Company

Arthur Holm has its origins in the Danish furniture designer Jorgen Alex Jensen, who was active during the sixties and the seventies. His design inspiration and his concept of ergonomics have been continued by his family, the design force behind Arthur Holm product range. The result of combining Scandinavian design tradition with Mediterranean creativity, flexibility and emotion is a wide product range built on more than 25 years of craftsmanship.

Arthur Holm offers a professional product range where tomorrow's technology is shaped into valued materials with design flexibility and customisation, specially created to enhance communication and collaboration in reception, conference and meeting areas.

The art of customisation

Arthur Holm offers a range of ingenious products whose designs are based on quality materials and the latest technology. Products which endow meeting and conference rooms with silent, ergonomic, innovative and aesthetic solutions that integrate into the furniture, hang from the walls as works of art or are used as interactive points of information.

The world of Arthur Holm

Arthur Holm offers a selection of unique, elegant, versatile, flexible and ergonomic products that are being used in meeting and conference rooms, reception areas, auditoriums and public zones of leading companies throughout the world.

The numbers speak for themselves! We currently own 33 product patents, have presence in over 45 countries with products and solutions in more than 25,000 installations.

It will be our pleasure to work with you, designing your unique and personalised environment. Our broadcast electronic engineers will provide the latest technology while our design team will offer you the most exclusive appearance.

Our team puts its heart, passion and pride in all our designs.

REGULATIONS AND SECURITY



ATTENTION: Do not disassemble or modify the device in any way. This symbol warns of the presence of dangerous un-insulated voltages inside some of the components, of sufficient magnitude to expose people to risk of electronic shock.



This symbol draws attention to important use and maintenance instructions in the manual that accompanies the unit.



This symbol indicates that the equipment conforms to the norms established by the European Community.



This symbol indicates D.C. current.

SAFETY INSTRUCTIONS

Plugs

- Do not dismantle any part of the device power connector.
- Disconnect the power plug from the AC outlet when the device is not going to be used for an indefinite period of time.

Power and extensions cords

- Use the appropriate power cord with the correct plug type.
- Do not overload wall outlets or power cords.
- Make sure the total ampere passed through an extension cord does not exceed the maximum allowed by the cable used.
- Do not place anything on the power cord.
- Do not locate this product where a person may walk or trip over the cord.

SAFETY INSTRUCTIONS



Wiring connected to hazardous voltage requires installation by qualified personnel or the use of ready-made flexible cables.



For your security, your equipment must be connected to an electrical outlet with grounding connection protection.

Since the plug is used to disconnect the device, the operating electrical outlet must be in an easily accessible place.

Environment

- Install the equipment on an elevated, flat surface.
- Install the equipment in a ventilated area.
- Avoid exposing the equipment to:
 - Rain or water
 - Excessive heat, cold or humidity
 - Area exposed to direct sunlight
 - Dirty areas
 - Equipment generating strong magnetic fields
- Avoid placing open containers of liquid, near the equipment.
- Keep a minimum distance of 30 mm in order to have good ventilation.
- Never place above the device any sources of flames such as lighted candles, etc.
- If you are using the device in extreme weather conditions and/or tropical climates, the equipment should be installed in a room which ensures a reasonable level of temperature and humidity.

SAFETY INSTRUCTIONS

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy, and if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

SYSTEM DESCRIPTION

General

ERT interfaces provide an easy and convenient system to control and diagnose motorised monitors, microphone lifts and other automated solutions by Arthur Holm helping to reduce the installation costs in terms of both equipments and time.

Easy to install and set-up, they facilitate a quicker installation and eliminate the need of external relay boxes and extra power outlets. It is also possible to have Arthur Holm's virtual support on site as long as there is an Internet connection available.

AHlink wireless control is also integrated in ERT interfaces, providing an easy and direct access to the most common settings through an app and making the start-up and future interventions hassle free. This means there is no need to access physically the unit to set-up or diagnose its installation.

Highlights

- OTAS (One Touch Addressing System)
- IP to RS422/RS485 interface
- ERT-60 2 AHnet inputs (RS-422) / ERT-30 1 AHnet input
- Controls up to 60 units (ERT-60)
- RDM system (Remote Diagnostic and Maintenance)
- Remote/local control and management
- Wireless control through AHlink app

Box Contents

Before the installation of your device please check the contents of the shipping box, it must contain the following items:

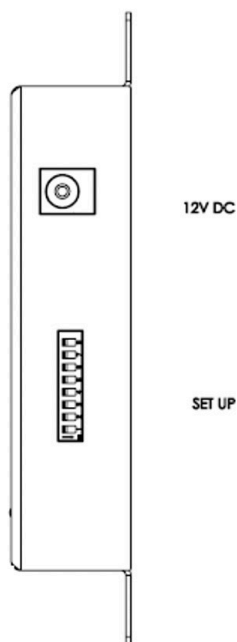
- ERT unit
- Power cord
- Power supply 100-2 40Vac, 50-60Hz. Output 12V
- User guide

SYSTEM DESCRIPTION

Available Models

MODEL	DESCRIPTION
ERT-30	IP to RS-422 interface to control, configure and diagnose ARTHUR HOLM motorised products. 1 AHnet port (RS422/ RS485) to control up to 30 units. Ethernet RJ45. OTAS (One Touch Addressing System). Remote/local control and management. RDM system (Remote Diagnostic and Maintenance). Wireless control through AHlink app.
ERT-60	IP to RS-422 interface to control, configure and diagnose ARTHUR HOLM motorised products. 2 AHnet ports (RS422/RS485) to control up to 60 units. Ethernet RJ45. OTAS (One Touch Addressing System). Remote/local control and management. RDM system (Remote Diagnostic and Maintenance). Wireless control through AHlink app.

CONTROL AND CONNECTORS



12V D.C.: Power input connector. Plug the power supply supplied.

SETUP: Eight deep switches to control and setup the unit.

DIP1: Activates the radio signal to control the unit using AHlink APP

DIP2: Activates the automatic AHnet ID configuration on the RS422 bus

DIP3: Not used

DIP4: Activates the firmware upgrade

DIP5: Not used

DIP6: Configures the factory IP address

IP: 192.168.0.128

MASK: 255.255.255.0

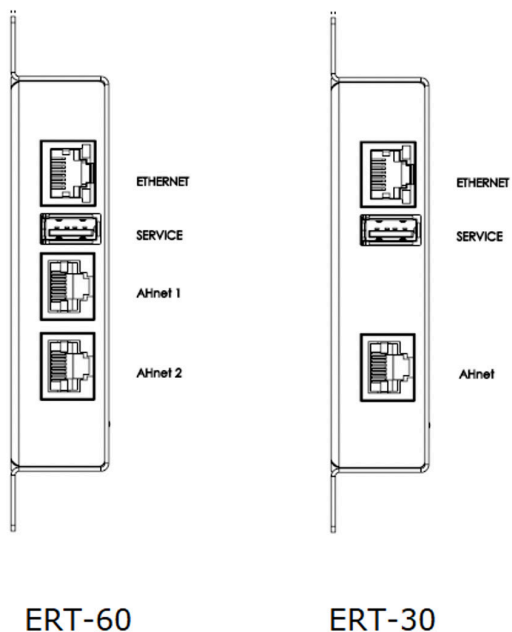
GATEWAY: 192.168.0.3

MAC: 70:B3:D5:49:70:00

DIP7: Not used

DIP8: Not used

CONTROL AND CONNECTORS



ETHERNET: RJ45 network connector. Connect the ERT to your local network, to open the 2002 port and send AHnet commands to the units connected to the RS422 ports (AHnet1 and AHnet2).

SERVICE: USB-A connector used only to firmware upgrade.

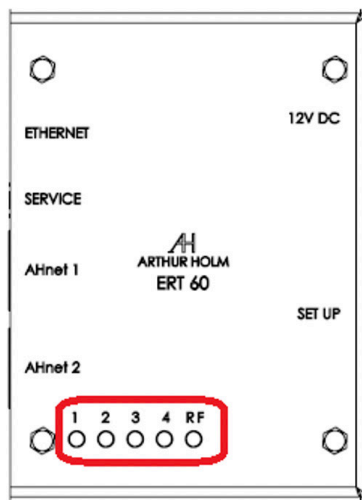
AHnet1: RJ45 RS422 port. Connect the Arthur Holm or Albiral units to control using AHnet commands.

AHnet1: RJ45 RS422 port. Connect the Arthur Holm or Albiral units to control using AHnet commands.

Note: ERT-30 has only one AHNet RS422 port connector. With an ERT-60, you can control up to 60 devices using the two AHNet port connectors (30 devices connected on each port).

CONTROL AND CONNECTORS

LEDs system indicators



ETHERNET: RJ45 network connector. Connect the ERT to your local network, to open the 2002 port and send AHnet commands to the units connected to the RS422 ports (AHnet1 and AHnet2).

LED1: Change the state when an AHnet command is received from the two RS422 ports (AHnet1 or AHnet2).

LED2: Change the state when the unit receives data from the local network port (ETHERNET).

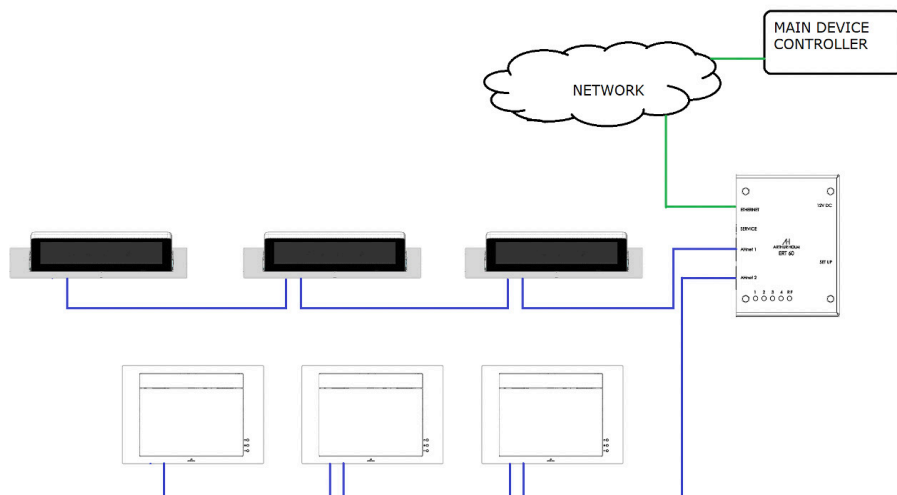
LED3: Not used.

LED4: Change the state when the unit send an AHnet command when the automatic AHnet ID configuration is activated (DIP2). The ERT interface sends commands to the units connected on AHnet1 and AHnet2 to automatic configure the AHnet ID on each unit.

RF: Brighths when a mobile or tablet is connected to the unit to control and configure the ERT using the AHlink APP.

SYSTEM CONFIGURATION

Use the ETHERNET unit port to connect the ERT to your local network area. Using this port, opens the 2002 ERT IP port, and send AHnet commands to control the devices connected to RS422 ERT ports (AHnet1 and AHnet2). The Arthur Holm or Albiral units, has an AHnet IN and AHnet OUT port to easily install a RS422 BUS. Next figure shows a schematic connection diagram.



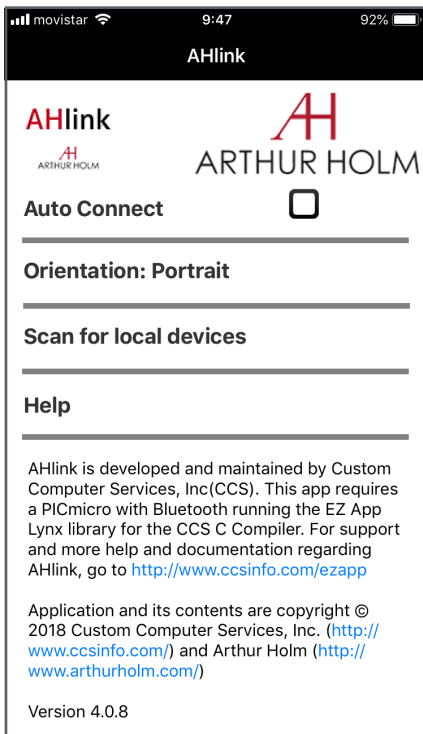
The main device controller connects to the ERT interface using the local network area. Just opening the 2002 ERT port, the main device controller can send AHnet commands to the ERT interface, and the ERT sends the AHnet commands to the units by the RS422 ports.

You can connect up to 60 units to the ERT using the two RS422 RJ45 buses (30 units on each bus). Please, follow the respective user manual to configure the AHnet address on each unit and to connect the unit to the RS422 bus.

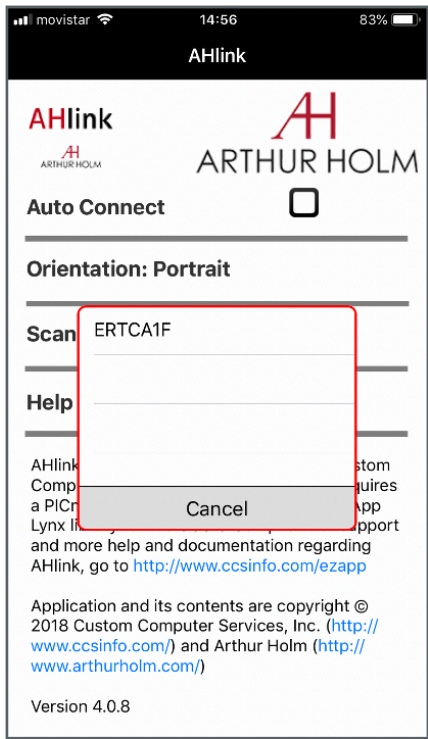
SYSTEM CONFIGURATION

To configure the ERT IP address, do you need to download the AHlink APP. First, do you need to activate the radio signal. Activate DIP1 on SETTINGS device, and power the ERT unit. To control and configure the unit using AHlink APP, download "AHLink" from App Store (IOS system) or from Google Play (Android system). The App is free. Execute the "AHLink" on your mobile phone.

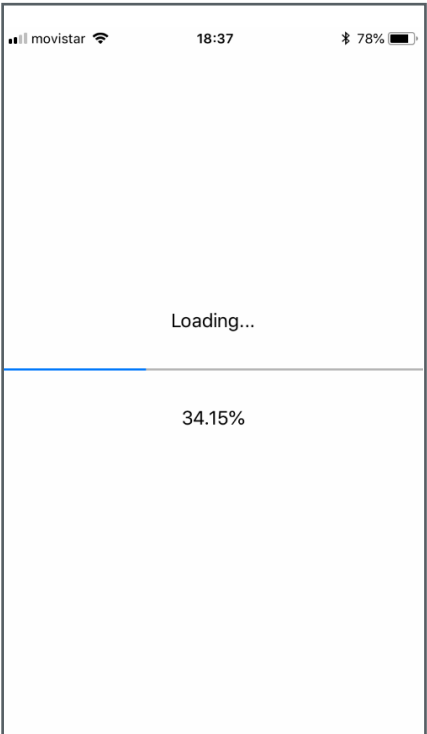
On App options, is recommended not select "Auto Connect" and select "Portrait" on "Orientation". To connect to the device, selects "Scan for local devices"



SYSTEM CONFIGURATION



The name always begins by “ERT” characters and the fourth last MAC radio address. Select the device to control and configure the unit.



Selecting the device, the first page is upload on the mobile phone. This page is used to check the device. When the mobile APP is connected to the unit the RF LED will bright.

SYSTEM CONFIGURATION

movistar

14:56

82%

<AHlink

Arthur Holm

ERT

Serial: 0000400915

Ver: 0.8

Address (0:Broadcast)

0

Dynamic2 Control

MOV

UP

DOWN

DISP

ON

OFF

BUT

LOCK

UNLOCK

IN

DVIA

DVI1

DVI2

Dynamic3 Control

MOV

UP

DOWN

DOWN F_UP

DISP

ON

OFF

BUT

LOCK

UNLOCK

IN

DVIA

DVI1

DVI2

Dynamic Talk Control

MOV

UP

DOWN

MIC

TALK

MUTE

BUT

LOCK

UNLOCK

IP Address

192.168.0.128

MASK

255.255.255.0

Gateway

192.168.0.3

Configuration PSW:

0

SERIAL

Serial number of the unit

VER

Firmware version

ADDRESS

Configure the AHnet Address to send the command. Use address 0 to broadcast the command instruction

Dynamic2 CONTROL

Send basic commands to Dynamic2 monitors family. Select the AHnet address to send the command, and press the function to send to AHnet RS422 BUS

Dynamic3 CONTROL

Send basic commands to Dynamic3 monitors family. Select the AHnet address to send the command, and press the function to send to AHnet RS422 BUS

DynamicTalk CONTROL

Send basic commands to Dynamic talk microphone family. Select the AHnet address to send the command, and press the function to send to AHnet RS422 BUS

IP ADDRESS: Shows the actual ERT IP address. Note that if the DIP6 is active, the unit is configured with the factory IP and this IP information should be different

MASK: Shows the actual ERT MAK address

GATEWAY: Shows the actual ERT gateway address

CONFIGURATION PSW: Enter 4006 password to configure the ERT IP address. Next page shows the ERT network configuration

SYSTEM CONFIGURATION

Network configuration

Change the network configuration and press "NETWORK INI" button to update the new network settings

movistar14:5882%

< AHlinkArthur Holm

ERT

Serial: 0000400915

Network configuration

IP[1]	192
IP[2]	168
IP[3]	0
IP[4]	128

MASK[1]	255
MASK[2]	255
MASK[3]	255
MASK[4]	0

Gateway[1]	192
Gateway[2]	168
Gateway[3]	0
Gateway[4]	3

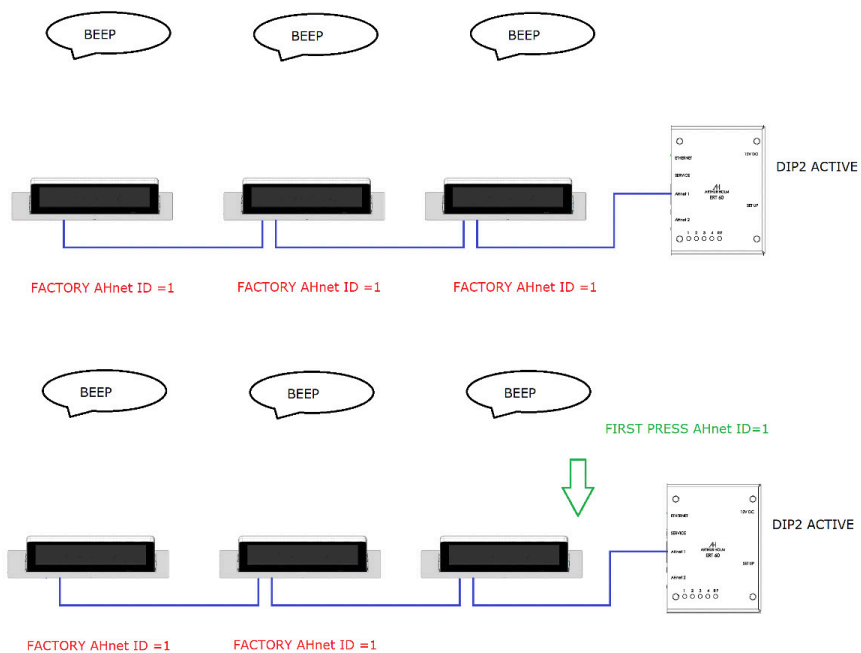
NETWORK INI

AHnet ADDRESS CONFIGURATION

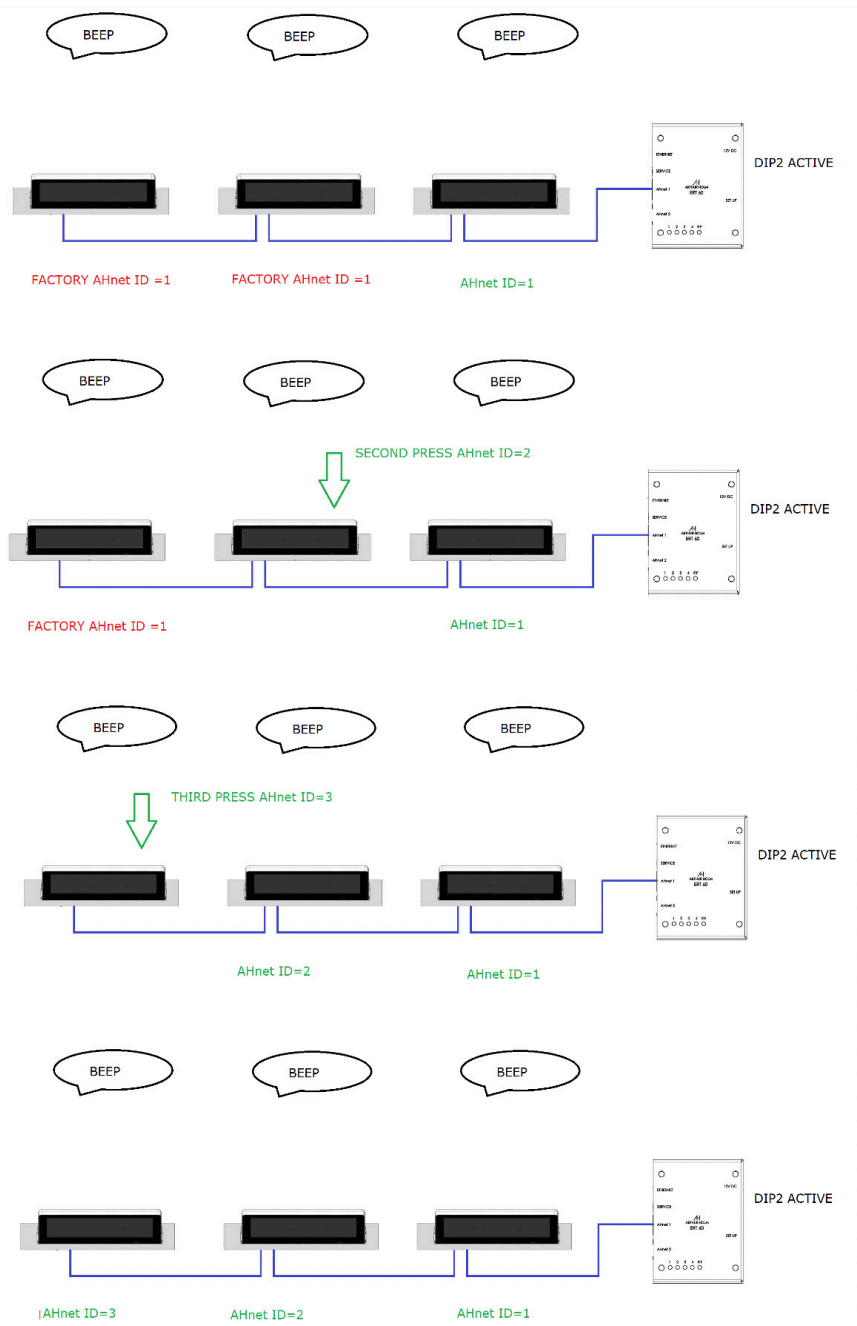
There is an ERT functionality that helps the users to configure the AHnet address. To send AHnet commands to a unit, first do you need to configure the AHnet address to individually send commands. Installing the unit (monitor, microphone or other device), do you need to configure a different AHnet address on each unit. The ERT has a utility that helps the assignation of the AHnet address task.

First of all, note that this utility it is only available on devices that has LCD screen configuration. The devices that has AHnet address rotary switches, don't have this functionality.

To activate this function, activate DIP2 on SETTINGS ERT switches, and power the unit. The ERT will periodically send a special AHnet command to configure the AHnet address on each unit. When the units receive the special AHnet address configuration command, you hear a BEEP sound. Pressing the UP button on unit cover plate, the unit configures her new AHnet address. The configuration is incremental. First unit that the user pres the UP button, gets AHnet address 1. Second, AHnet address 2...



AHnet ADDRESS CONFIGURATION



AHnet PROTOCOL

AHnet protocol

Uses 5 bytes communication:

BYTE 0	START BYTE
BYTE 1	ADDRESS BYTE
BYTE 2	COMMAND BYTE
BYTE 3	VALUE 1
BYTE 4	VALUE 2

BYTE 0 is the header and should always FA in HEX mode

BYTE 1 is the unit AHnet address to send the command. To send the command to all the devices, there are special AHnet address used as broadcast. Please, refer to the user manual unit as there are different family broadcast addresses

BYTE 2 is the function to execute on the unit

BYTE 3 first value of the function

BYTE 4 second value of the function. In some functions this value is not used and should be "00"

Below, there are some AHnet commands used to control the different family devices. You will find more AHnet commands on the particular user manual.

AHnet PROTOCOL

Dynamic2 family

COMMAND	DESCRIPTION	RESPONSE
FA XX 01 01 00	GO UP	FB XX 01 01 00
FA XX 01 00 00	GO DOWN	FB XX 01 00 00
FA XX 02 01 00	SCREEN ON	FB XX 02 01 00
FA XX 02 00 00	SCREEN OFF	FB XX 02 00 00
FA XX 03 01 00	DVI-A SELECTION	FB XX 03 01 00
FA XX 03 00 00	DVI 1 SELECTION	FB XX 03 00 00
FA XX 03 03 00	DVI 2 SELECTION	FB XX 03 03 00
FA XX 04 01 00	BUTTON LOCK	FB XX 04 01 00
FA XX 04 00 00	BUTTON UN LOCK	FB XX 04 00 00
FA XX 15 00 00	ASK BACKLIGHT LEVEL	FB XX 15 00 ZZ
FA XX 15 01 ZZ	SET BACKLIGHT LEVEL	FB XX 15 01 ZZ
FA XX 16 00 00	ASK CONTRAST LEVEL	FB XX 16 00 ZZ
FA XX 16 01 ZZ	SET CONTRAST LEVEL	FB XX 16 01 ZZ
FA XX 17 00 00	ASK BRIGHTNESS LEVEL	FB XX 17 00 ZZ
FA XX 17 01 ZZ	SET BRIGHTNESS LEVEL	FB XX 17 01 ZZ

ZZ Value (0-99Dec)

XX AHnet address

BROADCAST XX=F9

AHnet PROTOCOL

Dynamic3 family

COMMAND	DESCRIPTION	RESPONSE
FA XX 01 01 00	GO UP	FB XX 01 01 00
FA XX 01 00 00	GO DOWN CLOSED	FB XX 01 00 00
FA XX 01 02 00	GO DOWN SCREEN UP	FB XX 01 02 00
FA XX 02 01 00	SCREEN ON	FB XX 02 01 00
FA XX 02 00 00	SCREEN OFF	FB XX 02 00 00
FA XX 03 01 00	DVI-A SELECTION	FB XX 03 01 00
FA XX 03 00 00	DVI 1 SELECTION	FB XX 03 00 00
FA XX 03 03 00	DVI 2 SELECTION	FB XX 03 03 00
FA XX 04 01 00	BUTTON LOCK	FB XX 04 01 00
FA XX 04 00 00	BUTTON UN LOCK	FB XX 04 00 00
FA XX 15 00 00	ASK BACKLIGHT LEVEL	FB XX 15 00 ZZ
FA XX 15 01 ZZ	SET BACKLIGHT LEVEL	FB XX 15 01 ZZ
FA XX 16 00 00	ASK CONTRAST LEVEL	FB XX 16 00 ZZ
FA XX 16 01 ZZ	SET CONTRAST LEVEL	FB XX 16 01 ZZ
FA XX 17 00 00	ASK BRIGHTNESS LEVEL	FB XX 17 00 ZZ
FA XX 17 01 ZZ	SET BRIGHTNESS LEVEL	FB XX 17 01 ZZ

ZZ Value (0-99Dec)

XX AHnet address

BROADCAST XX=F8

AHnet PROTOCOL

DynamicTalk

COMMAND	DESCRIPTION	RESPONSE
FA XX 01 01 00	GO UP	FB XX 01 01 00
FA XX 01 00 00	GO DOWN	FB XX 01 00 00
FA XX 02 01 00	MUTE (PA)	FB XX 02 01 00
FA XX 02 00 00	TALK (PA)	FB XX 02 00 00
FA XX 04 01 00	BUTTONS LOCK	FB XX 04 01 00
FA XX 04 00 00	BUTTONS UNLOCK	FB XX 04 00 00

ZZ Value (0-99Dec)

XX AHnet address

BROADCAST XX=F8

INFORMATION ON DISPOSAL FOR USERS OF WASTE ELECTRICAL & ELECTRONIC EQUIPMENT



This symbol on the products and/or accompanying documents means that used electrical and electronic products should not be mixed with general household waste.

For proper treatment, recovery and recycling, please take these products to the designated collection points, where they will be accepted on a free of charge basis. Alternatively, in some countries, you may be able to return your products to your local retailer upon the purchase of an equivalent new product.

Disposing of this product correctly will help save valuable resources and prevent any potential negative effects on human health and the environment, which could otherwise arise from inappropriate waste handling. Please contact your local authority for further details of your nearest designated collection point.

Penalties may be applicable for incorrect disposal of this waste, in accordance with national legislation.

For business users in the European Union

If you wish to discard electrical and electronic equipment, please contact your dealer or supplier for further information.

Information on disposal in countries outside the European Union

This symbol is only valid in the European Union. If you wish to discard this product, please contact your local authorities or dealer and ask for the correct method of disposal.

WARRANTY TERMS AND CONDITIONS

Albiral Display Solutions warrants this product against manufacturing defects and workmanship for a period of two (2) years from the date of purchase, subject to the conditions below.

1. Electrical, electronic boards, accessories and power supply are warranted against manufacturing defects and workmanship for a period of two (2) years from the date of purchase
2. Labour costs: Albiral Display Solutions covers the labour costs to replace any defective parts during the validity of this warranty.
3. Transport costs:
 - 5.1. In the case that a manufacturing defect occurs within 90 days after the purchase date, both freight and insurance costs will be paid by Albiral Display Solutions.

Although Albiral Display Solutions S.L. pays for freight and insurance costs, Albiral Displays Solutions will not be responsible for any damages caused by the transportation of the goods if the customer does not inform in writing when receiving the goods.

- 5.2. After 90 days of the purchase date, the beneficiary of the warranty will pay both freight and insurance costs.

Albiral Display Solutions S.L. will not be responsible for any damages caused by transportation, when this one is paid by the customer.

4. This warranty does not cover the labour costs of handling, diagnose, removal, replacement, reinstall and/or program any product.
5. This warranty does not apply if the fault has been caused by misuse, improper handling, electrical or mechanical abuse, abnormal operation conditions, non-authorised modifications, and inadequate transportion or as a result of atmospheric phenomenons.

WARRANTY TERMS AND CONDITIONS

6. The warranty is not valid if people other than authorised Service Centre have handled the product and if the product has been manipulated or warranty seals are removed or manipulated.
7. The beneficiary of the warranty will have to return the product with the original packaging to warranty there are no damages during transportation.
8. The warranty is not valid if the beneficiary of the warranty does not include a RMA form and a copy of the purchasing invoice.

Please note that laws vary from country to country, and the same provisions of this warranty may not apply to you.

If you have any doubts concerning the terms of this warranty, please contact:

business@albiral.com

+34 938 502 376

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e-mail: marketing@arthurholm.com

Patents

MU17180ES00	P27178ITEP
MU17301ES00	P27178RUPC
MU17322ES00	P27178USPC
MU17413ES00	P27284DEEP
MU17445ES00	P27284EPPC
MU17854ES00	P27284ESEP
MU17868DEPC	P27284GBEP
MU17868RUPC	P27284RUPC
P24821DEEP	P27284USPC
P24821ESEP	P27715ESEP
P24821GBEP	P28089DEEP
P24821USPC	P28089ESEP
P27178DEEP	P28089ITEP
P27178EPDV01	P28089USPC
P27178EPPC	P28090EP00
P27178ESEP	P31160ESES
	P31160PCES